AI-Powered Smart Lock System

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements for the Course of

CSA1108 - OBJECT ORIENTED ANALYSIS AND DESIGN

to the award of the degree of

BACHELOR OF TECHNOLOGY IN

COMPUTER SCIENCE AND ENGINEERING
(B.Tech Computer Science and BioScience)

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled **"AI-Powered Smart Lock System"** has been carried out by **Ragul E(192571032)** under the supervision of Sarasu.s and is submitted in partial fulfilment of the requirements for the current semester of the **B.Tech Computer Science and BioScience** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

We, **Ragul E(192571032)** of the **B.Tech Computer Science and BioScience** , SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled **"AI-Powered Smart Lock System"** is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place: CHENNAI

Date:

Signature of the Students with Name

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Individual Contribution Statement

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Contribution
Ragul E (192571032)	**Module 1: AI-Based Facial Recognition and Project Coordination**	Developed the facial recognition workflow, including face capture, face detection, feature extraction, face matching, and authentication.	Tested the recognition process using authorized and unauthorized users and analysed authentication performance.	**Introduction, Engineering Problem, AI-Based Facial Recognition Module, and System Design**	**33.33%**
Imthiaz MD (192521360)	**Module 2: Smart Lock & Access Control**	Developed the access-control workflow, including authentication decision processing, ESP32 control, servo actuation, LED indicators, buzzer, and display updates.	Tested lock/unlock operations, access decisions, indicators, buzzer responses, and display functionality.	**Literature Review, Smart Lock Design, Access Control, and Module 2 Documentation**	**33.33%**
Sakthivel.S (192524378)	**Module 3: Real-Time Security Analytics & Web Application**	Developed the access-event recording process, backend, database, dashboard, and real-time security analytics.	Tested event logging, database storage, dashboard functionality, access history, and overall system integration.	**Implementation, Testing, Results, Conclusion, Future Work, References, and Module 3 Documentation**	**33.34%**
Total	**Complete AI-Powered Smart Lock System Development**	—	—	—	**100%**

ABSTRACT

The **AI-Powered Smart Lock System** is developed to provide secure and intelligent access control by replacing conventional key-based authentication with modern digital technologies. The major contribution of this module is the implementation of **AI-Based Facial Recognition** for identifying and authenticating authorized users.

The proposed facial recognition module begins by capturing the user's face using a camera. The captured image is processed to detect the facial region, after which important and unique facial features are extracted. These extracted features are compared with the facial information of registered users stored in the system database. Based on the matching result, the system determines whether the person is authorized.

The complete workflow of the module consists of **Face Capture, Face Detection, Feature Extraction, Face Matching, and Authentication**. When a registered user's face is successfully recognized, the authentication result is sent to the Smart Lock System, allowing the next module to control access. If the face does not match any registered user, authentication is rejected.

This module improves the security and convenience of the overall Smart Lock System by reducing dependence on physical keys and enabling automated identity verification. The module is evaluated based on authentication accuracy, recognition response time, reliability, and its ability to distinguish authorized and unauthorized users. However, factors such as lighting conditions, facial angle, distance, and changes in appearance may affect recognition performance.

Overall, **Module 1: AI-Based Facial Recognition** provides the intelligent authentication component of the AI-Powered Smart Lock System and plays a major role in ensuring that only verified users are allowed to proceed with the access-control process.

Keywords

Artificial Intelligence, Facial Recognition, Face Detection, Feature Extraction, Face Matching, Authentication, Computer Vision, Smart Lock System.

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
AI	Artificial Intelligence	---
FR	Facial Recognition	---
CV	Computer Vision	---
ML	Machine Learning	---
CNN	Convolutional Neural Network	---
RGB	Red, Green, Blue	---
API	Application Programming Interface	---
IoT	Internet of Things	---
FPS	Frames Per Second	fps
ms	Millisecond	ms
s	Second	s
MB	Megabyte	MB
%	Percentage	%

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

The AI-Powered Smart Lock System is designed to provide secure and intelligent access control using modern technologies. The most important requirement of an intelligent access-control system is the ability to identify whether a person attempting to gain access is authorized.

Traditional authentication methods such as physical keys and passwords have several limitations. Keys can be lost, stolen, or duplicated, while passwords and PINs can be forgotten, shared, or observed by unauthorized persons. Therefore, biometric authentication provides an alternative method by using unique human characteristics for identity verification.

In this project, **AI-Based Facial Recognition** is used as the primary authentication mechanism. The module uses a camera to capture the user's facial image and processes the image through multiple stages, including face detection, feature extraction, and face matching.

The main workflow of the module is:

Face Capture → Face Detection → Feature Extraction → Face Matching → Authentication

The final authentication result is provided to the overall Smart Lock System for further access-control operations.

1.2 Need for the Module

The Smart Lock System requires an intelligent method to determine whether a person is authorized before access is granted. Traditional authentication mechanisms do not automatically identify the physical identity of a person.

The AI-Based Facial Recognition Module addresses this problem by:

- Automatically identifying users.
- Providing keyless authentication.
- Reducing dependence on physical keys.
- Comparing captured faces with registered user data.
- Rejecting unknown or unauthorized users.
- Supporting automated access-control decisions.

The module therefore acts as the **intelligent identity verification component** of the overall project.

1.3 Problem Statement

The engineering problem addressed in this module is the development of a reliable **AI-Based Facial Recognition System** capable of identifying and authenticating authorized users in real time.
The system must capture a user's face, detect the facial region, extract meaningful features, compare those features with registered user data, and generate an authentication result.
The system should also handle factors that may affect facial recognition, including:

- Different lighting conditions.
- Facial angles.
- Distance from the camera.
- Facial appearance variations.
- Image quality.

The existing project already identifies lighting and environmental conditions as important challenges affecting facial recognition reliability.

1.4 Aim

To design and implement an AI-Based Facial Recognition Module that captures, detects, processes, matches, and authenticates user faces for secure identity verification in the AI-Powered Smart Lock System.

1.5 Objectives

- To capture the user's facial image using a camera.
- To detect the facial region from the captured image.
- To extract unique facial features for identification.
- To compare extracted features with registered user facial data.
- To identify authorized and unauthorized users.
- To generate an authentication decision based on the face-matching result.
- To provide the authentication result to the overall Smart Lock System.
- To evaluate recognition accuracy and response time.
- To analyse factors affecting facial recognition performance.

1.6 Scope

Included

The module includes:

- Face capture.
- Face detection.
- Facial image processing.
- Feature extraction.
- Face matching.
- User authentication.
- Registered user facial-data management.
- Authorized-user recognition.
- Unauthorized-user rejection.
- Authentication-result generation.

Excluded

The following are not the primary responsibility of this module:

- Electronic lock control.
- RFID authentication.
- Fingerprint authentication.
- PIN authentication.
- Door sensor management.
- Buzzer and alert hardware control.
- Remote web or mobile monitoring.

These functions belong to the other modules of the overall Smart Lock System.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Research related to:

- AI-based facial recognition.
- Face detection.
- Feature extraction.
- Face matching.
- Biometric authentication.
- Facial recognition in access-control systems.

2.2 Existing Approaches

The comparison will focus on:

Existing Approach	**Main Method**	**Advantage**	**Limitation**
Traditional Password/PIN	Knowledge-based authentication	Simple	Can be shared or observed
RFID Authentication	Card/tag-based access	Fast	Card can be lost
Fingerprint Recognition	Biometric authentication	Unique identity	Requires physical interaction
Basic Face Detection	Image-based detection	Contactless	Limited identification capability
AI-Based Facial Recognition	AI-based feature matching	Intelligent and contactless	Affected by environmental conditions

2.3 Engineering Gap

The main gap addressed by this module is the requirement for an intelligent, contactless authentication mechanism that performs complete:

Face Capture → Face Detection → Feature Extraction → Face Matching → Authentication

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS

3.1 Module Requirements

Functional Requirements

Requirement	Description
Face Capture	Capture the user's facial image
Face Detection	Identify the face from the image
Feature Extraction	Extract unique facial information
Face Matching	Compare with registered facial data
Authentication	Accept or reject the user

Non-Functional Requirements

Requirement	Expected Requirement
Accuracy	Reliable recognition under normal conditions
Response Time	Fast authentication
Usability	Simple user interaction
Reliability	Consistent recognition performance
Security	Protect registered facial information

3.2 Module Design

AI-BASED FACIAL RECOGNITION WORKFLOW

```
    USER
     |
     ▼
  FACE CAPTURE
     |
     ▼
  FACE DETECTION
     |
     ▼
 FEATURE EXTRACTION
     |
     ▼
  FACE MATCHING
     |
     ▼
  AUTHENTICATION
    |
    ▼
 AUTHORIZED  UNAUTHORIZED
```

3.3 Detailed Module Functions

A. Face Capture

The camera captures the facial image of the user when authentication is requested.

B. Face Detection

The captured image is analysed to identify and isolate the user's face from the background.

C. Feature Extraction

Important and unique facial characteristics are converted into meaningful features that can be used for identification.

D. Face Matching

The extracted facial features are compared with the features of registered users stored in the system.

E. Authentication

The system generates the final result:

- **Face Matched → Authenticated**
- **Face Not Matched → Authentication Rejected**

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS

4.1 Technologies Used

The report will focus on the technologies relevant to your module, such as:

Tool / Technology	**Purpose**
Camera / ESP32-CAM	Face capture
Python	AI and image-processing implementation
OpenCV	Face detection and image processing
AI Recognition Model	Feature analysis and recognition
Database / JSON	Registered facial-data storage

The original report also identifies Python, OpenCV, TensorFlow/Keras, camera hardware, and user facial-data processing as relevant technologies for the facial recognition implementation.

4.2 Testing

Test ID	**Function**	**Expected Result**
T01	Face Capture	Face image captured successfully
T02	Face Detection	User's face detected
T03	Feature Extraction	Facial features extracted
T04	Face Matching	Features compared with database
T05	Authorized User	Authentication accepted
T06	Unauthorized User	Authentication rejected

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The AI-Based Facial Recognition Module was developed as the intelligent authentication component of the AI-Powered Smart Lock System. The module performs face capture, face detection, feature extraction, face matching, and authentication to verify whether a user is authorized.

The module provides a contactless and automated method for identity verification. By comparing the captured user's facial information with registered user data, the system can generate an authentication decision that supports the overall access-control process.

The major contribution of the module is the development of an intelligent authentication workflow that reduces dependence on traditional keys and manual identity verification.

5.2 Future Work

Future improvements can include:

- Improved recognition under low-light conditions.
- Faster facial recognition models.
- Support for different facial angles.
- Anti-spoofing mechanisms.
- Improved privacy protection.
- Encrypted facial-data storage.
- Multi-factor authentication.
- Offline facial recognition.
- Support for a larger number of registered users.

5.3 Individual Reflection

E. Ragul (192571032)

My individual contribution to the AI-Powered Smart Lock System focused on **Module 1: AI-Based Facial Recognition**. My main responsibility was designing the authentication workflow consisting of face capture, face detection, feature extraction, face matching, and authentication.

Through this module, I gained knowledge about Artificial Intelligence, computer vision, facial recognition, image processing, feature extraction, and biometric authentication. I also understood the importance of recognition accuracy, response time, lighting conditions, and reliable authentication in a security-based system.

IMPORTANT CHANGE I WILL FOLLOW

Overall Project	**Your Individual Report**
AI-Powered Smart Lock System	Same overall project
Complete Smart Lock	**Module 1: AI-Based Facial Recognition**
All hardware functions	Only explain them when connected to your module
Electronic lock control	Mention only as the next system response
RFID/Fingerprint/PIN	Not your primary contribution
Facial recognition	**Your main contribution**
Your workflow	**Face Capture → Detection → Feature Extraction → Matching → Authentication**

The original report format already contains all five major chapters, appendices, testing sections, contribution evidence, and the overall Smart Lock System context. I will retain that **same report format** while changing the technical focus of every relevant section to your AI-Based Facial Recognition Module.